A Pilot Evaluation of a Virtual Reality Educational Game for History Learning

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Abstract: Several studies have suggested that the use of ICT may motivate students in history learning and help them develop historical thinking. Virtual and augmented reality, mixed reality and tangible environments and other similar technologies can provide authentic, interactive, and explorative experiences to the students, moving away from the traditional book-based education into new immersive game-based learning experiences. In this study, we present a virtual reality (VR) game for history learning, and the results of a pilot study with students. In the game, the students are moving around virtual Acropolis where six trials are waiting for them. Each trial is related with a Greek myth and students have to complete a number of different activities for each trial in order to proceed to the next one. The aim of the game is to complement third graders' history learning in schools. The game is free of charge, and the only infrastructure that is necessary is a typical smartphone and a Google Cardboard. A pilot study with twenty-eight (28) primary school students took place in the context of an exhibition focused on gamed-based learning. Data were collected through questionnaires and focus group discussions. Students' responses revealed their positive attitude towards the VR game since they considered it as simple, innovative, valuable, inspiring, challenging, practical, predictable and appropriate for learning about history. That's why they also supported that similar environments would have been of great value in schools. Students supported that they were fully focused on the tasks at hand and they felt present in the virtual environment.

Keywords: historical thinking, history learning, game-based learning, virtual reality

1. Introduction

For many centuries people have been trying to understand what it means to be a part of history and how and why particular people become historically important figures (Korallo, 2010). History satisfies man's instinct of curiosity about past developments in all aspects of life (Adesote and Fatoki, 2013) while history education provides students with knowledge about the past, how it has determined the present and the way we live. Although history education should promote students' critical thinking (Yilmaz, 2008), latest evidence suggest that traditional history teaching leads to sterile memorization of historical knowledge while students acquire a poor understanding of historical events and processes. Therefore, it is necessary to find new strategies that could help students get a better, broader and more comprehensive understanding of history (Howson, 2007). The recent prompts for teaching history focus not only on the knowledge that the students will acquire but also on the skills that they will develop and which will allow them to understand, analyze and interpret the facts. Students are encouraged not just to remember facts, but to familiarize themselves with the way historians work: they are asked to construct their own, personal, meaning about what the sources "are telling" them.

To this end, several studies suggest to employ technology for improving history teaching (Masterman and Rogers, 2002). Information and communication technologies (ICT) can be used to increase students' motivation, interest and enjoyable feelings during the learning process (Boadu et al., 2014) and transform history learning to an explorative and critical thinking approach (Blanco-Fernández et al., 2014). Many ubiquitous technologies such as augmented and virtual reality as well as tangible or mixed reality installations are continuously being examined in the history learning context (Triantafyllidou et al. 2017). These approaches offer interactive and explorative experiences to the students, moving away from the traditional book-based education into a new immersive game-based experience. Such approaches engage students much more than traditional learning and add to the learning experience higher entertainment levels. Korallo et al. (2012) suggest that virtual environments may be especially useful for history learning since they can provide interactivity and motivation. Korallo (2010) underlines that interactive virtual environments can allow students to actively control the educational environment, can give them the feeling that are present at a particular period of time in history, can provide easy access to some realistic historical materials and consequently can support more the understanding of diversity in history. Historical video games, virtual museums, augmented reality guidance into archeological sites are some well-known applications exploiting these technologies.

In this study, we present a virtual reality (VR) game for history learning, and the results of a pilot study with students. In this game, students take the role of an ancient Greek hero and visit different places, solve puzzles and answer interactive quizzes.

2. Virtual reality in education

Virtual reality (VR) is an interactive computer-generated experience taking place within a simulated environment. It uses computing power to create and simulate virtual environments from which the user has the illusion of being surrounded and to which he can move freely, interacting with the objects they include, as he would do in the real world.

Researchers and education stakeholders recognized virtual reality as a powerful tool for the learning process. VR can transform learners from passive information receivers to an active knowledge explorers. The introduction of virtual reality technology in primary, secondary and higher education began in the early 1990's with projects such as Science Space, Safety World, Global Change, Virtual Gorilla Exhibit, Atom World, and Cell Biology (Youngblut, 1998). However, limitations such as the installation cost (Merchant et al., 2014; Riva, 2003), headset weight and fit, simulator sickness (Costello, 1993) and poor instructional design of the virtual learning environments (Chen, Toh & Ismail, 2005), restricted the widespread dissemination of VR in education.

Despite the initial limitations, the rapid increase of the processing power, the reduction of VR technology infrastructure cost, the design of immersive 3D virtual environments and the high-speed internet connection increased again the prospects of VR technology in educational settings (Merchant et al., 2014). Free and easy to use authoring systems such as Blender and Unity 3d, for the development of 3D objects and 3D interactive environments, allow researchers and instructional designers to exploit VR technology in various domains and educational settings.

The particular features and affordances offered by virtual reality seem to contribute to positive learning outcomes (Mikropoulos & Natsis, 2011). Virtual reality allows to rebuild past worlds, monuments and ancient cities and enable the comparison of different historical periods and their physical transformation. VR technology has also been used to promote critical historical thinking and cultural heritage. Architecture and archaeology also explore this technology to enhance the understanding of cultural heritage (Maietti et al., 2017). Virtual museums take advantage of virtual reality in order to display, preserve, reconstruct and store collections in a digital form (Liarokapis et al., 2017). The ability of the users to take, manipulate, redistribute and redescribe digital objects of the past is considered as the primary educational value of virtual museums (Bayne et al., 2009).

There are a lot VR applications for history education. For example, the 3DMURALE project (Cosmas et al., 2001) developed 3D multimedia tools to record, reconstruct, encode and visualize archaeological sites such as the ancient city of Sagalassos in Turkey. The Foundation of the Hellenic World (FHW) has produced a number of VR applications for representing the Olympic Games in ancient Greece (Gaitatzes et al., 2004; Blanco-Fernández et al., 2014). A characteristic example is 'Walk through Ancient Olympia', where the user apart from visiting the historical site, learns about the ancient games themselves by interacting with non-player characters (NPC), such as athletes in the ancient game of pentathlon (Liarokapis et al. 2017). Koutsabasis and Vosinakis (2018) developed a VR kinesthetic application of sculpturing Cycladic figurines, which places the user at the role of an ancient craftsman who creates a figurine with bare hand movements. Eggarxou and Psycharis' (2007) used 3D environments to allow students to explore the Erechtheum in ancient Athens. There are also some VR games for Google Cardboard such as Acropolis VR (Mozaik Education, 2018) and Acropolis experience (Unimersiv, 2018).

3. Trials of the Acropolis

Most of the previously mentioned VR applications focus more on the realistic approximation of the archeological sites and they are less focused on the instructional interactions that promote historical thinking skills. In this study, we present a virtual reality game based on the world of Greek mythology aimed to function as a supportive tool for history lessons of 3rd grade students in Greek schools. The game is called "Trials of the

Acropolis", it has been developed with using free programs and technologies (Chintiades et al., 2017) and is available for free from Google app store (in Greek and English editions) (Chintiadis, 2018).

The game aims to familiarize students with five myths. The rationale of the game was to let the users participate actively in the learning process and learn through the acquisition of experience rather than through passive exploration of the virtual space. For this reason, emphasis was given on the game design.

The design of "Trials of the Acropolis" was based on the "Design, Play and Experience" (DPE) framework of Winn (2009). This framework proposes a methodology for designing serious games through three aspects: the *Design*, *Play* and *Experience*. Each of them has four common layers, namely, the Learning, Storytelling, Gameplay and User experience layer (Winn, 2009).

At the *Learning* layer the designer has to prepare the educational content and pedagogy elements in order to aid the game's learning purpose. The first step of this process is to consider the target audience and the learning outcomes of the end-user experience. Just as the teachers prepare their instructional interventions or guide the students during a course, the designer needs to define the learning outcomes using proven techniques.

The *Storytelling* layer provides a story with a different perspective for the designer and the player. The designer, with the help of the narrative, the setting and the character design tools, lays the core foundation of the main scenario having in mind to provide purpose and engagement. The player, however, experiences a different story because during gameplay the elements of the designer's story are combined with the interactions that the player performs, thus forming a different story experience.

The *Gameplay* layer is important because it represents the players' interactivity and what he is allowed to do within the game world. The mechanics, dynamics and affects are the main integral parts of this layer. Generally, the mechanics define the operations within the game. The dynamics with the contribution of the player's choices provide the resulting behavior. The affects are the overall experiences and emotions. The user *Experience* layer is generally what the player sees, experiences and hears during the game. For this reason, the designer must develop a truly engaging experience and story, with an easy to use interface so as to make the player focuses more on the gameplay and learning aspects rather than the interface complexities.

Based on the DPE framework we exploited a specific sequence of interactions for every user trial in the Acropolis: user engagement, narrative-based learning, hermeneutics, assessment as shown in Table 1.

Table 1: Aspects and steps of the game trials

Design	Play	Experience
Be acquainted with the archeological site	Student starts walking and staring at the virtual world	Engagement
Familiarize and understand myths through digital storytelling	In a specific and related point of the virtual world, a character narrates the myth to the students while presenting related images	Narrative-based Learning
Offer puzzles of related images that will help students better understand the myth	The student solves the virtual puzzle and discovers the meaning of the image	Game-based learning – Hermeneutics
Offer interactive quiz for consolidating the myth qualities	Student answers the interactive quiz in order to fulfill the trial and get feedback	Game-based learning – Self Assessment

Initially, the user is placed in a specific area of the Acropolis so as to explore the place, feel part of it and engage with the virtual world. More specifically, the player is placed at the entrance of Propylaea, visiting the Acropolis for the first time and without knowing anything about the challenges which he will confront. Afterward, he meets a character, an ancient warrior, that has to fulfill an ancient myth, which according to the scenario, no one in the history of Ancient Greece has ever managed to. The character narrates the myth with the use of an image slider synchronized with the narration (fig.1) and informs the user about the trial objectives. At the next step, the student has to solve a virtual puzzle (fig.2) that reveals an image related to the specific myth. Students are also asked to provide their interpretation of the image. Finally, at the end of every trial, the user has to answer an interactive quiz so as to assess and improve his understanding of the myth. The quiz is consisted of five multiple choice questions. Behind the user, there are five pictures that give information about the quiz questions. Therefore, the user can turn his head and study these images so as to consolidate his knowledge and answer the questions (Fig.3). After each response, appropriate constructive feedback is given. If the user answers correctly at least four out of five questions, he can continue on the next trial. Otherwise, he has to repeat this step.

The player come across six trials, "The trials of the Acropolis" as they called. The first five correspond to specific myths that are subjects of the 3rd grade history book while the last trial tries to revise all the previous myths. The myths that are addressed in the game are short versions of *Gods and Titans*, *Hercules*, *Theseus*, *Jason and the Argonauts*, *Odyssey*.

Figure 1: The avatar narrating a story

Figure 2: Solving the puzzle

Figure 3: Answering the quiz

4. Methodology

4.1 Purpose and research questions

The purpose of this study is to examine and describe the students' attitudes and experience towards the VR game describe previously.

The research questions were the following:

- Do the students assess the VR mobile game as interesting and engaging?
- Do the students consider the VR world as authentic and expressive?
- Do the students consider the VR mobile game as effective and usefull?

4.2 Participants and procedure

Trials of the Acropolis was explored by 28 primary school students, in the context of an exhibition focused on gamed-based learning for Primary Schools at the city of Kavala in Greece. Regular Android smartphones with earphones, google cardboards and Bluetooth controllers were used to play the game.

Initially, students were introduced to the scope of the game and the technology used. Afterwards, they wear the cardboards and the earphones and started playing the game (Fig. 4). During the game, they had to complete at least three trials. The game endured about 15 minutes. Upon the completion of the game, students participated in a attitude-questionnaire about their experience and participated in focus group discussions.

Figure 4: Students playing the game

4.3 Research Instruments

Data collection was based on an attitude-questionnaire and focus group discussions. The questionnaire consisted of 23 7-point Likert questions and evaluated the usability and attractiveness of the VR game and the student's experience. Most of the questionnaires' items were derived from AttrakDiff (Hassenzahl & Monk, 2010), Flow State Scale (Jackson & Marsh, 1996), and Reality Judgment and Presence Questionnaire (Baños et al. 2000). The following variables assessed the engagement, the learning effectiveness, and the authenticity of virtual experience :

- *Pragmatic Quality* (4 items): Measures the extent to which the system enables a user to achieve his goals (e.g. *the environment was simple – complicated*);
- *Hedonic Quality-Stimulation* (3 items): Measures the extent to which the system is perceived as innovative and interesting (e.g. *the environment was conservative - innovative*);
- *Hedonic Quality-Identity* (3 items): Measures the extent to which the system lets the user to identify with it (e.g. *the environment was cheap - valuable*);
- *Autotelic experience* (3 items): Measures the extent to which the system offers user fulfillment (e.g. *the experience left me feeling great*);
- *Perceived learning* (3 items): Measures students' perceptions about the educational value of the system (e.g. *I would prefer to learn about history with similar environments*);
- *User Focus* (3 items): Measures the students' perceived focus on the learning activities during system usage (e.g. *I was completed focused on task*);
- *Reality judgment* (4 items): Measures the willingness to interpret virtual experiences as if they were veridical (e.g. *To what extent did your interactions with the virtual world seem natural to you, like those in the real world*)

All variables can be considered as consistent since they had satisfactory Cronbach's α as seen in Table 2.

The focus group discussions were conducted upon the completion of the questionnaires in groups of two students and aimed at capturing their qualitative views about the learning environment. The questions were focused on what they enjoyed and disliked and their perceptions in regards to the learning value and the learning efficiency of the environment.

5. Results

5.1 Attitude-questionnaire

As seen in Table 2, students' general assessment of their learning experience with the presented game was positive. Students' responses in the mini AttrakDiff questionnaire revealed that they considered the game as simple, practical, predictable and appropriate for learning about history. Most students also supported that the environment was stimulating, innovative, valuable and appealing without moments of boredom and discomfort (hedonic quality). Students' answers also showed that they identified themselves with it (Hedonic Quality-Identity) and they thought that it offered inspiring, novel and challenging functions and interactions (Hedonic Quality-Stimulation). Students' answers in the last variable was higher in comparison to the other variables.

Table 2. Students' attitudes towards the presented educational VR game

	Min	Max	Mean	SD	Cronbach's a
Pragmatic Quality	2.00	7.00	5.03	1.61	.71
Hedonic Identity	1.00	7.00	5.06	1.85	.91
Hedonic Stimulation	1.00	7.00	5.56	1.76	.91
Focus	2.00	7.00	5.71	1.30	.87
Autotelic Experience	1.67	7.00	5.82	1.47	.89
Perceived learning	2.00	7.00	5.32	1.68	.86
Reality Judgment	3.25	7.00	5.50	1.14	.77

The virtual reality app managed to retain their attention for the entire game duration. Students supported they were focused on the tasks to be done (M= 5.82, SD=1.3). They were also delighted with their experience (autotelic experience) and, for example, they supported that the experience left them feeling great (M= 5.64, SD=1.87). Most students indicated that they would prefer to learn about history with similar environments (M= 5.30, SD= 1,83) and with that way they would also learn faster (M= 5.48, SD=1.89).

Students also pinpointed that they felt like being in the places presented. A sense of presence in a virtual environment stems from feeling as if you existed within but as a distinct entity from a virtual world that also exists (Baños et al. 2000). For example, students felt as part of the virtual world (M= 5.96, SD=1.26) and considered themselves as active participants of the narrated story (M= 5.56, SD=1.22). This is quite significant since games work only if people feel that they are real (Baños et al. 2000).

When we examined Spearman's correlation between reality judgement variable and the rest of the variables (none of which followed the normal contribution), the reality judgement variable seemed to be correlated significantly with every other variable except Hedonic Identity. For example, it was related with focus ($r= .70$, $p<.001$) and autotelic experience ($r= .69$, $p<.001$) and also perceived learning ($r= .45$, $p<.05$). That means that making learning environment more real was crucial for its success.

5.2 Focus group discussions

Students' comments validated their answers in the questionnaires. At the beginning of the game, the students looked excited since most of them had never any experience with Google Cardboard and a VR game. They stated repeatedly that they were impressed with the 3D virtual environment of the Acropolis. Their comments confirmed the illusion of being in the place, the illusion of self-embodiment and the illusion of physical interaction, hence, the students fell like being a part of the virtual world.

> *"It was a great experience to move accross the ancient Acropolis, watching Parthenon and meet up ancient warriors. "*
>
> *"I thought that it could be real. It was impressive."*
>
> *"I had played other 3D games in my console. However the use of cardboard make me fill as it was real... I felt that I was part of this ancient world."*
>
> *"What I enjoyed more was moving around Acropolis and staring at the buildings and statues..."*

Students were also positive in regards to the trials design, the different steps required to be completed. Several of them mentioned that completing a trial gave them satisfaction and the feel of success. Both the quiz and the puzzle steps were considered as interesting and competitive.

> *"I feel great when completing a trial..."*
>
> *"The quiz questions where intriguing and I am happy because I answered all of them and completed the trials."*

The instructional design managed to hide formal learning processes under the umbrella of the VR game. Students commented that the game "Trials of the Acropolis" helped them learn or remember the myths that were part of the trials. Students supported that they learnt these myths in an enjoyable way. They also underlined that similar games for other learning domains would be more than welcome.

> *"I wasn't thinking that this was about learning. For me it was like any other role playing game."*
>
> *"I wish we could learn things this way. I could play, I mean study, for hours."*
>
> *"I learnt details about the myths that I didn't know till now. It would be great if other sections of the book could also be covered by this or a new VR game."*

When students were informed that this game was available for free at the Google play store and the low cardboard cost, they claimed that they will ask their parents to buy a cardboard and use their phones in order to play this game again and explore other VR games. Students comments seemed to confirm that VR educational games could be used for history learning.

6. Discussion and conclusion

Virtual reality enables students to explore worlds they never thought possible. There are a lot of virtual reality apps aiming at advancing museum's learning experience but there are less focusing on learning about history in traditional classrooms. In this paper, we proposed a virtual reality application focusing on historical skills and historical understanding, and we also presented the results a pilot study evaluating students' experience with it. Students' responses revealed their positive attitude towards the VR game since they considered it as simple, innovative, valuable, inspiring, challenging, practical, predictable and appropriate for learning about history. That's why they also supported that similar environments would have been of great value in schools. Students were focused when using the game and they felt present in the virtual environment. The current findings extend prior research in virtual environments and serious games and indicate that students can get acquainted with history in a playful manner using VR technology and game based learning. However, we have to underline that the learning experience had a short duration and that means that the interactive learning content was also limited. Therefore, such applications can function as complimentary activities in the context of a more integrated instructional intervention.

The proposed environment is based on the design framework of DPE (Winn, 2009), and exploits discovery learning, storytelling, game-based learning and self-assessment. Our study provides evidence that their combination may provide an efficient and effective learning experience for history learning. The consecutive execution of tasks such as navigating to the virtual world, interacting with digital storytelling, solving virtual puzzles and answering questions, worked as expected. However, there are a lot to be done in order to create a coherent framework of designing virtual reality apps for history learning.

The storification opportunities (Akkerman, et al 2009) of the virtual reality applications are many with different qualities. Storification refers to structuring, and simultaneously making sense of experiences. Storification is a means to combine, episodes, actors, actions, and accounts of actions in time and space (Akkerman, et al 2009). In this context, the designers have to select whether they want the students to become a) receivers of historical narratives in virtual worlds without having influence on the content or the structure of the plot b) constructors of the historical events by defining and organizing the narrative elements, by building the virtual world or c) participants in the historical narrative by acting out specific roles and influence the historical progress. The last two types of interactions are less usual and are the ones matching the needs of developing historical understanding. We aim at advancing the proposed virtual reality app in order to include all three types of storification modes.

Our study has several limitations. The most important one is that we present the perceived learning value of the students. Although their views are a good indicator, they cannot offer definite answers for the learning effectiveness of the virtual reality app. Additionally, we do not analyze thoroughly the different phases of the underlying learning mechanism, their separate contribution to the learning result and students' experience. These issues are the target of future research with the Trials of the Acropolis app.

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